

AMATH 403 HW 7

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Question 1: 8.1.4

(a) Use the Fourier transform to solve the initial value problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, x) = \delta'(x - \xi), \quad -\infty < x < \infty, \quad t > 0,$$

whose initial data is the derivative of the delta function at a fixed position ξ .

$$\begin{aligned} u_t &= u_{xx} \\ \mathcal{F}u_t &= \frac{\partial \hat{u}}{\partial t} \\ \mathcal{F}u_{xx} &= (ik)^2 \hat{u} = -k^2 \hat{u} \end{aligned}$$

Form and solve the ODE

$$\begin{aligned} \frac{\partial \hat{u}}{\partial t} &= -k^2 \hat{u} \\ \frac{\partial \hat{u}}{\hat{u}} &= -k^2 \partial t \\ \ln(\hat{u}) &= -k^2 t + A(K) \\ \hat{u}(t, k) &= A(k) e^{-k^2 t} \end{aligned}$$

Transform initial condition

$$\begin{aligned} \hat{f}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx \\ \hat{u}(0, k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta'(x - \xi) e^{-ikx} dx \end{aligned}$$

$$f(x) = e^{-ikx}, \quad f'(x) = -ike^{-ikx}$$

$$\begin{aligned} \hat{u}(0, k) &= \frac{1}{\sqrt{2\pi}} \left(-(-ike^{-ik\xi}) \right) = \frac{ik}{\sqrt{2\pi}} e^{-ik\xi} \\ \hat{u}(0, k) &= A(k) e^0 = A(k) \\ A(k) &= \frac{ik}{\sqrt{2\pi}} e^{-ik\xi} \\ \hat{u}(t, k) &= \frac{ik}{\sqrt{2\pi}} e^{-ik\xi} e^{-k^2 t} \end{aligned}$$

Use inverse fourier transform

$$\begin{aligned}\hat{u}(t, k) &= ik \left(\frac{1}{\sqrt{2\pi}} e^{-ik\xi} e^{-k^2 t} \right) \\ \hat{v}(t, k) &= \frac{1}{\sqrt{2\pi}} e^{-ik\xi} e^{-k^2 t} \\ v(t, k) &= \mathcal{F}^{-1} \left(\frac{1}{\sqrt{2\pi}} e^{-ik\xi} e^{-k^2 t} \right) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-\xi)^2}{4t}}\end{aligned}$$

Since ik means take the spacial derivative:

$$\begin{aligned}u(t, x) &= \frac{\partial}{\partial x} v(t, x) \\ u(t, x) &= \frac{\partial}{\partial x} \left(\frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-\xi)^2}{4t}} \right) \\ u(t, x) &= \frac{1}{4\pi t} \left(e^{-\frac{(x-\xi)^2}{4t}} \right) \left(-\frac{x-\xi}{2t} \right) \\ u(t, x) &= -\frac{x-\xi}{4\sqrt{\pi t^3/2}} e^{-\frac{(x-\xi)^2}{4t}}\end{aligned}$$

(b) Show that your solution can be written as the derivative $\partial F/\partial x$ of the fundamental solution $F(t, x; \xi)$. Explain why this observation should be valid.

$$\begin{aligned}F(t, x; \xi) &= \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-\xi)^2}{4t}} \\ u(t, x) &= -\frac{x-\xi}{4\sqrt{\pi t^3/2}} e^{-\frac{(x-\xi)^2}{4t}}\end{aligned}$$

Take the spacial derivative of F

$$\begin{aligned}\frac{\partial F}{\partial x} &= \frac{1}{\sqrt{4\pi t}} \left(-\frac{x-\xi}{2t} \right) \left(e^{-\frac{(x-\xi)^2}{4t}} \right) = -\frac{x-\xi}{4\sqrt{\pi t^3/2}} e^{-\frac{(x-\xi)^2}{4t}} \\ u(t, x) - \frac{\partial F}{\partial x} &= \left(-\frac{x-\xi}{4\sqrt{\pi t^3/2}} e^{-\frac{(x-\xi)^2}{4t}} \right) - \left(-\frac{x-\xi}{4\sqrt{\pi t^3/2}} e^{-\frac{(x-\xi)^2}{4t}} \right) = 0\end{aligned}$$

$\frac{\partial F}{\partial x}$ is a solution to the heat equation, and it fits the initial condition $u(0, x) = \delta'(x-\xi)$ because the heat equation is linear with constant coefficients. Because solutions to the initial value problem for the heat equation are unique, $\frac{\partial F}{\partial x}$ must be the exact same function as our solution $u(t, x)$.

Question 2: (8.1.17) Find the fundamental solution for the cable equation $v_t = \gamma v_{xx} - \alpha v$ on the real line.

Fundamental solution means

$$v(0, x) = \delta(x)$$

We can make the guess $v(t, x) = e^{-\alpha t} u(t, x)$

$$v_t = (-\alpha e^{-\alpha t} u) + e^{-\alpha t} (u_t)$$

$$v_t = e^{-\alpha t} (u_t - \alpha u)$$

$$v_x = e^{-\alpha t} u_x$$

$$v_{xx} = e^{-\alpha t} u_{xx}$$

Substitute back into ODE

$$e^{-\alpha t} (u_t - \alpha u) = \gamma (e^{-\alpha t} u_{xx}) - \alpha (e^{-\alpha t} u(t, x))$$

$$u_t - \alpha u = \gamma u_{xx} - \alpha u$$

$$u_t = \gamma u_{xx}$$

It looks just like the heat equation

$$v(0, x) = u(0, x)$$

$$\delta(x) = u(0, x)$$

$u(t, x)$ is the fundamental solution to the heat equation.

$$u(t, x) = \frac{1}{\sqrt{4\pi\gamma t}} e^{-\frac{x^2}{4\gamma t}}$$

$$v(t, x) = e^{-\alpha t} \left(\frac{1}{\sqrt{4\pi\gamma t}} e^{-\frac{x^2}{4\gamma t}} \right)$$

$$v(t, x) = \frac{1}{\sqrt{4\pi\gamma t}} e^{-\left(\frac{x^2}{4\gamma t} + \alpha t\right)}$$

Question 3: (8.3.5) Consider the parabolic equation

$$\frac{\partial u}{\partial t} = x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}, \quad 1 < x < 2,$$

with initial and boundary conditions

$$u(0, x) = f(x), \quad u(t, 1) = \alpha(t), \quad u(t, 2) = \beta(t).$$

(a) State and prove a version of the Maximum Principle for this problem.

If $u(t, x)$ is a smooth solution to $u_t = xu_{xx} + u_x$ on the domain $R = \{(t, x) : 0 \leq t \leq T, 1 \leq x \leq 2\}$ then the maximum value of u is at a point on the boundary of R .

We will prove this by contradiction. We can start by assuming $u(t, x)$ has its maximum value at some interior point $u(t_0, x_0)$. Since it is the maximum value, the derivatives must be equal to zero.

$$u_t(t_0, x_0) = 0$$

$$u_x(t_0, x_0) = 0$$

And the second derivative must be less than or equal to zero.

$$u_{xx}(t_0, x_0) \leq 0$$

We can plug in our values at (t_0, x_0) in to our ODE:

$$u_t(t_0, x_0) = xu_{xx}(t_0, x_0) + u_x(t_0, x_0)$$

$$0 = x(\leq 0) + 0$$

This simplifies down to $0 \leq 0$, which is true but does not provide a full contradiction.

We can define $v(t, x)$ which is perturbed:

$$v(t, x) = u(t, x) - \epsilon t$$

where $\epsilon > 0$ is an arbitrarily small number. Since $u(t, x) = v(t, x) + \epsilon t$, we can find the derivatives:

$$u_t = v_t + \epsilon$$

$$u_x = v_x$$

$$u_{xx} = v_{xx}$$

Subbing these into the original PDE gives us the PDE for v :

$$v_t + \epsilon = xv_{xx} + v_x \implies v_t = xv_{xx} + v_x - \epsilon$$

Now we can assume $v(t, x)$ has its maximum at some interior point (t_0, x_0) . The spatial derivatives are zero and less than 0:

$$\begin{aligned}v_x(t_0, x_0) &= 0 \\v_{xx}(t_0, x_0) &\leq 0\end{aligned}$$

For the time derivative, since we are defining our domain up to time T , v_t could be zero positive if the maximum happens right at the very end of the time limit T .

$$v_t(t_0, x_0) \geq 0$$

We can substitute these values at (t_0, x_0) into our new ODE for v :

$$\begin{aligned}v_t(t_0, x_0) &= x_0 v_{xx}(t_0, x_0) + v_x(t_0, x_0) - \epsilon \\v_t(t_0, x_0) &= x_0(\leq 0) + 0 - \epsilon\end{aligned}$$

Since our domain is $1 \leq x \leq 2$, we know x_0 is a strictly positive number. A positive number multiplied by ≤ 0 is still ≤ 0 . This leaves us with:

$$v_t(t_0, x_0) \leq -\epsilon$$

Since ϵ is a positive number, $-\epsilon$ is negative, which means $v_t(t_0, x_0)$ must be strictly less than zero. but we already established that $v_t(t_0, x_0) \geq 0$.

We have $0 \leq v_t \leq -\epsilon < 0$, which is a contradiction.

Because of this contradiction, our assumption is false. So $v(t, x)$ cannot have its maximum on the interior, and must have it on the boundary. If we let $\epsilon \rightarrow 0$, $v(t, x)$ turns back into $u(t, x)$, which means $u(t, x)$ must also have its maximum on the boundary.

(b) Establish uniqueness of the solution to this initial-boundary value problem.

Let's assume we have two solutions, $u_1(t, x)$ and $u_2(t, x)$, that both solve the PDE and the given initial and boundary conditions.

We can define a new functions $w(t, x)$ such that:

$$w(t, x) = u_1(t, x) - u_2(t, x)$$

The PDE is linear, so we can subtract the PDE for u_2 from the PDE for u_1 , which shows us that $w(t, x)$ will also solve the PDE:

$$w_t = xw_{xx} + w_x$$

Now we can check the initial and boundary conditions for w . Since u_1 and u_2 share the exact same conditions, they will subtract to zero:

$$\begin{aligned}w(0, x) &= u_1(0, x) - u_2(0, x) = f(x) - f(x) = 0 \\w(t, 1) &= u_1(t, 1) - u_2(t, 1) = \alpha(t) - \alpha(t) = 0 \\w(t, 2) &= u_1(t, 2) - u_2(t, 2) = \beta(t) - \beta(t) = 0\end{aligned}$$

This means that on the entire boundary, the value of $w(t, x)$ is 0.

Using the maximum principle, the maximum value of $w(t, x)$ must be on this boundary. So $w(t, x) \leq 0$ everywhere on the interior domain.

We can also use the maximum principle on $-w(t, x)$ to show that the minimum value of $w(t, x)$ must also be on the boundary.

Since the boundary is 0, this means $w(t, x) \geq 0$ everywhere on the interior.

Since $0 \leq w(t, x) \leq 0$, $w(t, x) = 0$ everywhere on our domain.

Therefore, $u_1(t, x) - u_2(t, x) = 0 \implies u_1(t, x) = u_2(t, x)$ which shows that any two solutions to the PDE must be identical, so the solution is unique.